

The Open Methodology MoBI Laboratory

Nicole Burke, PhD; Nathalia Bianchini Esper, PhD; Bryan Gonzalez, PhD; Apurva Gokhe; Alp Erkent, MD; Iktae Kim; Camilla Strauss; Celia Maiorano; Freymon Perez; John Vito d'Antonio-Bertagnolli; Stan Colcombe, PhD; Alexandre Franco, PhD; Michelle Freund, PhD; Michael P. Milham, PhD

Child Mind Institute, New York, NY; USA

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Introduction

The past 25 years has seen an explosion in multimodal brain imaging studies (Warbrick, 2022). A multimodal approach allows researchers to integrate complementary data types and provide richer insights into brain function, structure, and connectivity (Calhoun & Sui, 2016; Wagner et al., 2019). A disadvantage to this approach is that each modality (EEG, eye-tracking, physiological measures) is often captured on separate devices, each with their own hardware and software. This makes data collection burdensome, needing to operate multiple separate devices, and introduces challenges in data consolidation because data needs to be temporally synchronized.

Here, we present a multimodal brain/body imaging laboratory (MoBI) that streamlines data collection into one centralized computer, using Lab Streaming Layer (LSL) (Kothe et al., 2024). This approach reduces hardware complexity, ensures real-time monitoring, and maintains the temporal fidelity of multimodal datasets.

Methods

Multimodal data collection presents two challenges: recording distinct data modalities across different devices and integrating time-synced data modalities after data collection. Our approach solves both of these problems. We developed a centralized system for data collection that allows for all data modalities to be directly connected to and recorded on a single computer, while using LSL. LSL uses a shared system clock for recordings, which allows for precise synchronization and effective correction for time drift across devices, eliminating the need for post hoc adjustments.

To implement our system, we developed documentation that provides detailed, step-by-step guidance on how to set-up a single computer MoBI system – everything from [equipment evaluation to data collection](#). The documentation includes specifications for the

necessary hardware components to build a MoBI system, insights into the configuration of our centralized computer, and open-source software built for this approach.

Results

We have successfully integrated a range of data acquisition systems into our single-computer MoBI framework, including dry and semi-dry EEG systems; tabletop and wearable eye tracking devices; and physiological measurement modalities such as electrocardiography, electrodermal activity, inductive respiration, and peripheral oxygen saturation. Additionally, the system supports multiple audio and video streams, motion capture systems, and cognitive task presentations presented on PsychoPy and Curious (Klein et al., 2021). Figure 1 compares our setup with a traditional MoBI laboratory.

By using LSL, our data is consolidated into a single file at time of acquisition. This has two major benefits. First, it alleviates the burden of needing to consolidate data post-data collection; the data recorded in LSL is already temporally synchronized. Second, it allows for rapid assessment of data quality control using a single open-source tool to quickly batch process recordings, assessing for major deviations from protocol length, data modality availability, and more.

Conclusions

Preliminary testing shows the LSL framework's robust capability to achieve clock synchronization across diverse devices, ensuring accurate temporal alignment of multimodal datasets. Our detailed documentation, publicly hosted at childmindresearch.github.io/MoBI_Docs provides the research community with a valuable resource for replicating and adapting these methodologies for multimodal studies. This documentation was used to replicate the lab five times, both domestically and internationally, with external collaborators in South Africa and Brazil.

Figure 1

Multi-computer systems versus our single-computer MoBI system

A. Traditional MoBI set-up



B. Single-Computer MoBI set-up



Note. Photo A depicts a traditional MoBI laboratory set-up, where each different modality needs its own computer and display for simultaneous data collection. Photo B shows our novel single-computer system. This single computer can support simultaneous data collection for EEG, eye-tracking, physiological measures, audio, video, motion tracking, and behavioral measures recorded on an iPad.

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